Computer Networking For Devops Engineer

How does the internet work? - The Internet works with optical fibres / fibres cable which connect us through a data centre where all the data gets stored and transmit it through cables.
Internet service providers installs the router, wifi and other devices for us and through this router and broadbands an IP address gets generated and identifies the devices to send the data.

DATA CENTER ——---> CABLES—---------> CABLES PROVIDER——----> INTERNET SERVICE PROVIDER(ISP)------------------------> GENERATE THE IP ADDRESS TO IDENTIFIES THE DEVICES——--> SEND DATA

Internet Work

What is a server ? - A server is a computer program or device that provides a service to another computer program and its user, also known as the client.

Server - To Provide the information.
Client - To Request the information.

Example -

YouTube —--------->	DNS (Server) ——---->	IP Address of server
(Domain)	Links each other	(Connect to particular Server)

Facebook.com —--------> ISP (checks the internet connection)---------> Optical fibre flows the data information ——---> data centre has that application which helps to stream in your mobile.

Difference between web servers and application servers -

How a web server works

A web server is technology that hosts a website's code and data. When you enter a URL in your browser, the URL is actually the address identifier of the web server.

Your browser and web server communicate as follows:

1. The browser uses the URL to find the server's IP address
2. The browser sends an HTTP request for information
3. The web server communicates with a database server to find the relevant data
4. The web server returns static content such as HTML pages, images, videos, or files in an HTTP response to the browser
5. The browser then displays the information to you

A website that hosts static content like blogs, header images, or articles can run on a web server. However, most websites and web applications are much more interactive and require an application server.

How an application server works

An application server extends the capabilities of a web server by supporting dynamic content generation, application logic, and integration with various resources.

When you attempt to access interactive content on a website, the process works as follows:

1. The browser uses the URL to finds the server's IP address
2. The browser sends an HTTP request for information
3. The web server transfers the request to the application server
4. The application server applies business logic and communicates with other servers and third-party systems to fulfil the request
5. The application server renders a new HTML page and returns it as a response to the web server
6. The web server returns the response to the browser
7. The browser displays the information to you

To use the example of an ecommerce website, when you add items to your cart, or check out items, you interact with the application server.

Types of Applications:

1) **Standalone Application** - A standalone application, is a software program designed in such a way that to run this software program, users don't need an internet connection or any server access.

2) **Web Application -** Web-based applications need an internet connection, servers, and any additional resources to run.
   Suppose you want to connect with a web application you need multiple servers. If you want to connect data stored applications then it requires a database server where all the data gets stored to send.

What is application maintenance? -

Application maintenance is the continuous process of updating, modifying, and re-assessing the existing software application in order to make it bug-free or improve performance. Application maintenance must be a constant task that ensures the application is running smoothly without any errors.

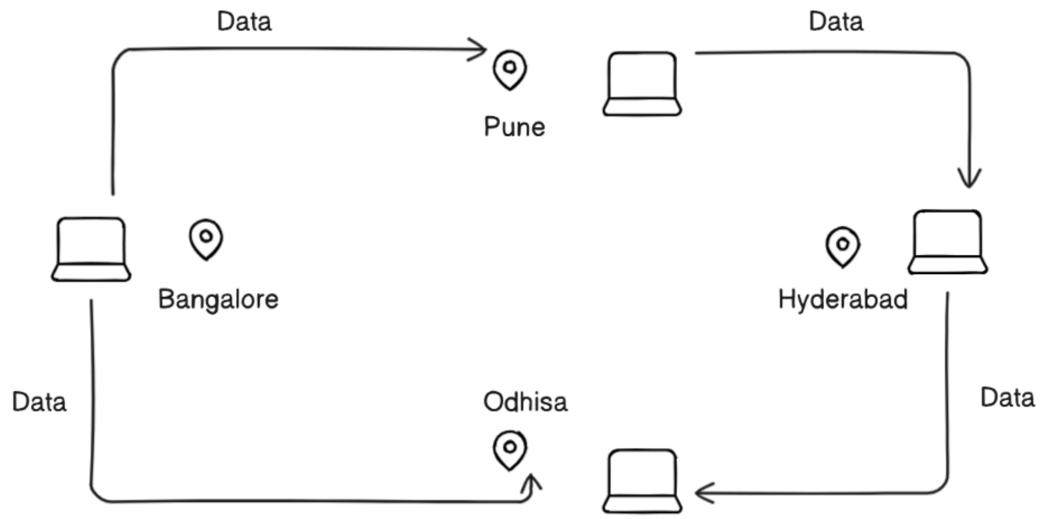
Application Support - Applications support usually offers technical support and human support. Technical support relates to the maintenance of the app, software and other organisation's technological systems. Human support relates to helping the end-user understand how to use the application or system and help solve any issues.

What is Data Center - Data Center is like a place where lots of computers are placed and stores all the data and transmits the store data with the help of cables and stays in the sea.

What is computer networking?

Computer networking is like having a group of friends who all have phones and can call or text each other. In computer networking, instead of phones, we have computers and instead of phone lines, we use cables, Wi-Fi, or other methods to connect them. When computers are connected to a network, they can share information and resources, like files, printers, and internet connections. This allows them to communicate with each other quickly and easily, just like friends talking on their phones.

It is a group of nodes or servers that are connected via a network that might be an internet is called computer networking.



A computer network is a system that connects many independent computers to share information (data) and resources. The integration of computers and other different devices allows users to communicate more easily. A computer network is a collection of two or more computer systems that are linked together. A network connection can be established using either cable or wireless media. Hardware and software are used to connect computers and tools in any network.

Why Networking is Important For DevOps Engineer -

Computer networking is crucial for Devops engineers, it acts as a backbone for communication between different computers or systems allowing them to manage and automates applications deployments across the environments, transmit the resources from one to another region, ensure security, troubleshoot network issues, optimize performances to maintain smooth operations.

OSI Model - The OSI (Open Systems Interconnection) model is a framework that describes how data moves from one device to another over a network. It divides the process into 7 layers, each with a specific role. Here's a simple breakdown:

1. Physical Layer

- **What it does**: Deals with the physical connection (wires, cables, signals).
- **Think of it as**: The hardware that sends and receives raw bits (like 1s and 0s).
- **Example**: Ethernet cables, Wi-Fi signals.

2. Data Link Layer

- **What it does**: Packages raw bits into frames and handles error detection/correction for reliable data transfer.
- **Think of it as**: Traffic control, ensuring smooth communication between devices on the same network.
- **Example**: MAC addresses, Ethernet.

3. Network Layer

- **What it does**: Routes data between different networks (choosing the best path).
- **Think of it as**: A GPS for your data, finding the way from sender to receiver.
- **Example**: IP addresses, routers.

4. Transport Layer

- **What it does**: Ensures reliable delivery of data by splitting it into smaller chunks and reassembling it.
- **Think of it as**: A delivery service that guarantees your package arrives intact.

- **Example**: TCP (reliable), UDP (fast but not guaranteed).

5. Session Layer

- **What it does**: Manages sessions (connections) between applications.
- **Think of it as**: The host keeping track of a conversation, ensuring it stays organized.
- **Example**: Logging into a website and keeping the session alive.

6. Presentation Layer

- **What it does**: Formats and encrypts data for the application layer.
- **Think of it as**: A translator that ensures data is readable and secure.
- **Example**: Data compression, encryption, file formats (like JPEG or MP3).

7. Application Layer

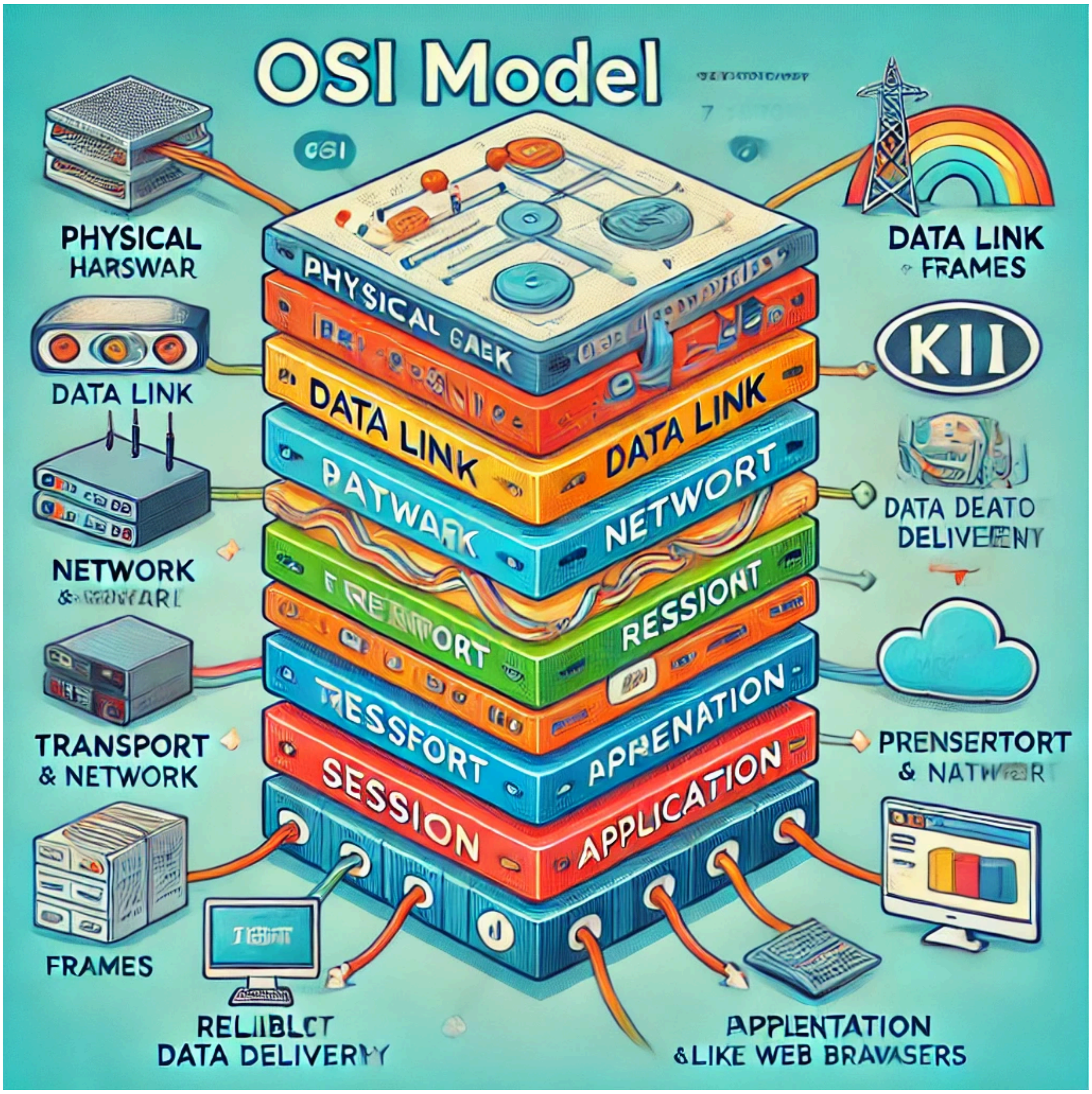
- **What it does**: Interfaces directly with the end-user and applications.
- **Think of it as**: The apps you use, like web browsers or email clients.
- **Example**: HTTP, FTP, DNS.

Summary Analogy

Imagine sending a letter:

- **Physical Layer**: The mailbox and delivery truck.
- **Data Link Layer**: Ensures the envelope isn't damaged.
- **Network Layer**: Finds the best route to the destination.
- **Transport Layer**: Ensures all pages arrive together.
- **Session Layer**: Keeps the communication open during the process.
- **Presentation Layer**: Formats the letter in a readable way.
- **Application Layer**: You write and read the letter.

OSI Diagram -



TCP/IP Model - The TCP/IP model is like the backbone of how the internet works. It describes how data travels between computers and devices on a network. Unlike the OSI model's 7 layers, the TCP/IP model has 4 layers. Here’s an easy way to understand it:

1. Application Layer

- **What it does**: Handles the apps and services you use (like web browsers, email, or video calls).
- **Think of it as**: The user's interaction with the internet.
- **Example protocols**: HTTP (for websites), FTP (for file transfers), SMTP (for email).

2. Transport Layer

- **What it does**: Makes sure data is delivered reliably and in the correct order.
- **Think of it as**: A delivery service ensuring your package arrives safely.
- **Protocols**:
    - **TCP (Transmission Control Protocol)**: Reliable but slower. Ensures all pieces of data arrive correctly. (Like sending a package with tracking and a receipt).
    - **UDP (User Datagram Protocol)**: Fast but less reliable. Good for real-time stuff like video or gaming. (Like sending a quick text without worrying about delivery confirmation).

3. Internet Layer

- **What it does**: Decides the path data takes to reach its destination, like a GPS.
- **Think of it as**: The navigator that finds the best route between networks.
- **Example protocol**: **IP (Internet Protocol)**, which assigns addresses (like your home address) so devices know where to send data.

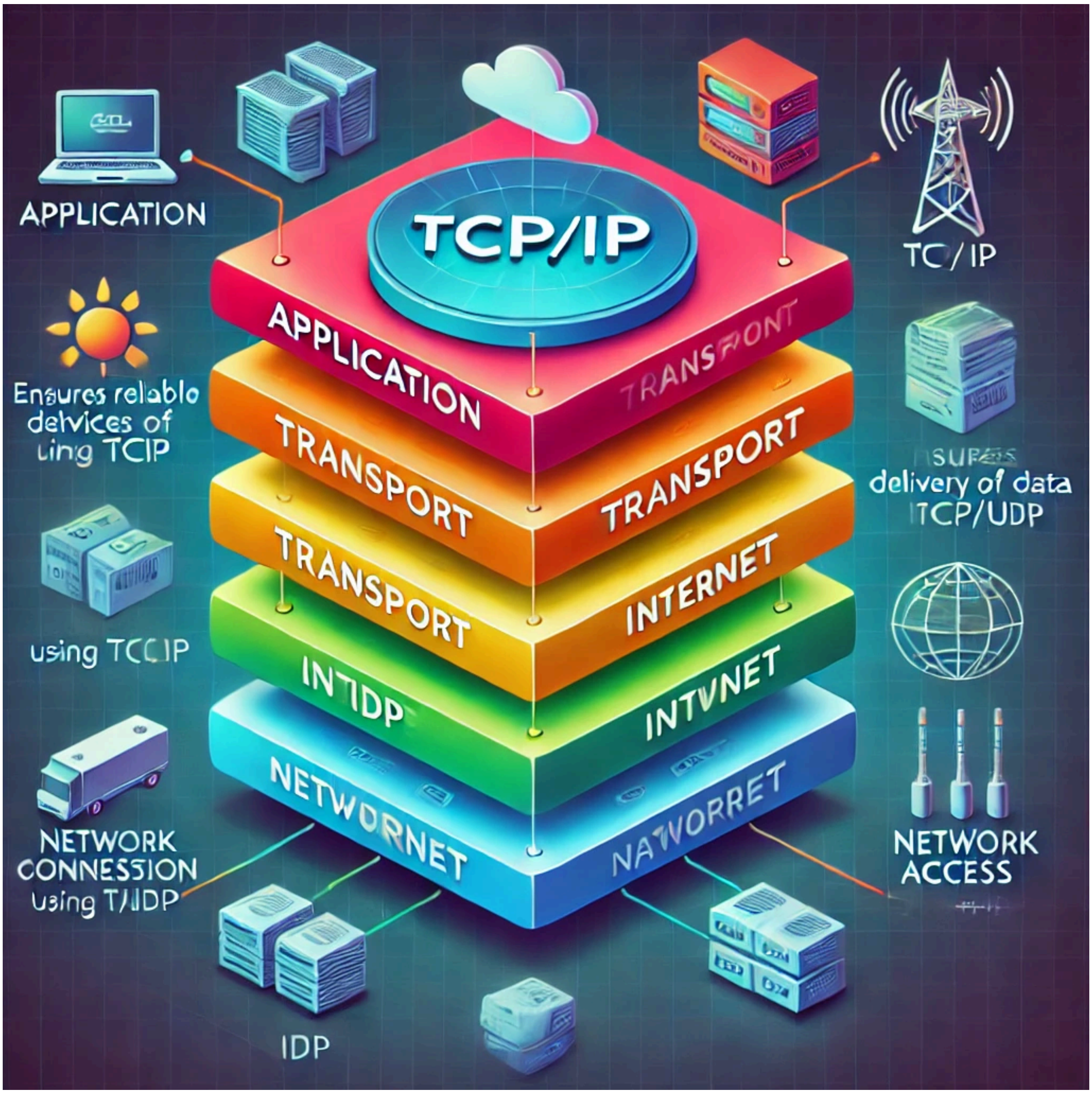
4. Network Access Layer (or Link Layer)

- **What it does**: Deals with the physical connection to the network.
- **Think of it as**: The hardware and signals that carry your data over cables or Wi-Fi.
- **Example technologies**: Ethernet, Wi-Fi.

Analogy: Sending a Letter

- **Application Layer**: Writing the letter (e.g., "Visit our website!").
- **Transport Layer**: Splitting it into pieces, ensuring each arrives (like tracking a package).
- **Internet Layer**: Finding the address and the best route (e.g., the recipient's IP address).
- **Network Access Layer**: Sending the envelope via the postal service (actual cables, signals).

TCP/IP Diagram -



IP (Internet Protocol) - An IP address, or Internet Protocol address, is a unique string of numbers assigned to each device connected to a computer network that uses the Internet Protocol for communication. It serves as an identifier that allows devices to send and receive data over the network, ensuring that this data reaches the correct destination.

Types of IP Address

IPV4 - The Internet Protocol version 4 (IPv4) address is the older of the two, which has space for up to 4 billion IP addresses and is assigned to all computers. IPV4 addresses are limited. It contains 32 bits.

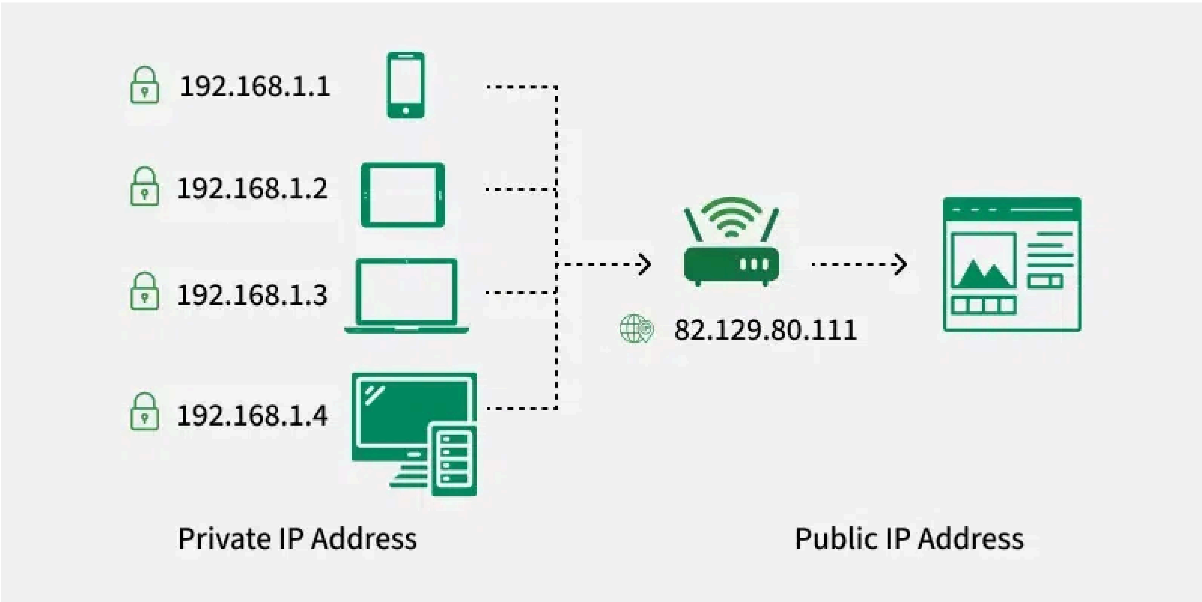
IPV6 - The more recent Internet Protocol version 6 (IPv6) has space for trillions of IP addresses, which accounts for the new breed of devices in addition to computers. It contains 128 bits.

Public IP Addresses - Public IP addresses are typically used for servers hosting websites, email servers, or any device that needs to be accessible from the internet. For instance, if you host a website on your own server at home, your ISP must assign a public IP address to your server so users around the world can access your site.

A Public IP address is assigned to every device that directly accesses the internet. This address is unique across the entire internet.

Private IP Addresses - Private IP addresses are used within private networks (such as home networks, office networks, etc.) and are not routable on the internet.

In a typical home network, the router assigns private IP addresses to each device (like smartphones, laptops, smart TVs) from the reserved ranges. These devices use their private IPs to communicate with each other and with the router. The router uses NAT to allow these devices to access the internet using its public IP address.



What is a Subnet?

A subnet is like a smaller group within a large network. It is a way to split a large network into smaller networks so that devices present in one network can transmit data more easily. For example, in a company, different departments can each have their own subnet, keeping their data traffic separate from others. Subnet makes the network faster and easier to manage and also improves the security of the network.

Why Is Subnetting Necessary?

- Subnetting helps in organizing the network in an efficient way which helps in expanding the technology for large firms and companies.
- Subnetting is used for specific staffing structures to reduce traffic and maintain order and efficiency.

- Subnetting divides domains of the broadcast so that traffic is routed efficiently, which helps in improving network performance.
- Subnetting is used to increase network security.

Example - VPC - Virtual Private Cloud.

DNS (Domain Name System) -

The Domain Name System (DNS) is like the internet's phone book. It helps you find websites by translating easy-to-remember names (like www.example.com) into the numerical IP addresses (like 192.0.2.1) that computers use to locate each other on the internet. Without DNS, you would have to remember long strings of numbers to visit your favorite websites. It is an application layer protocol for message exchange between clients and servers. It is required for the functioning of the Internet.

What is the Need for DNS?

Every host is identified by the IP address but remembering numbers is very difficult for people. Also the IP addresses are not static therefore a mapping is required to change the domain name to the IP address. So DNS is used to convert the domain name of the websites to their numerical IP address.

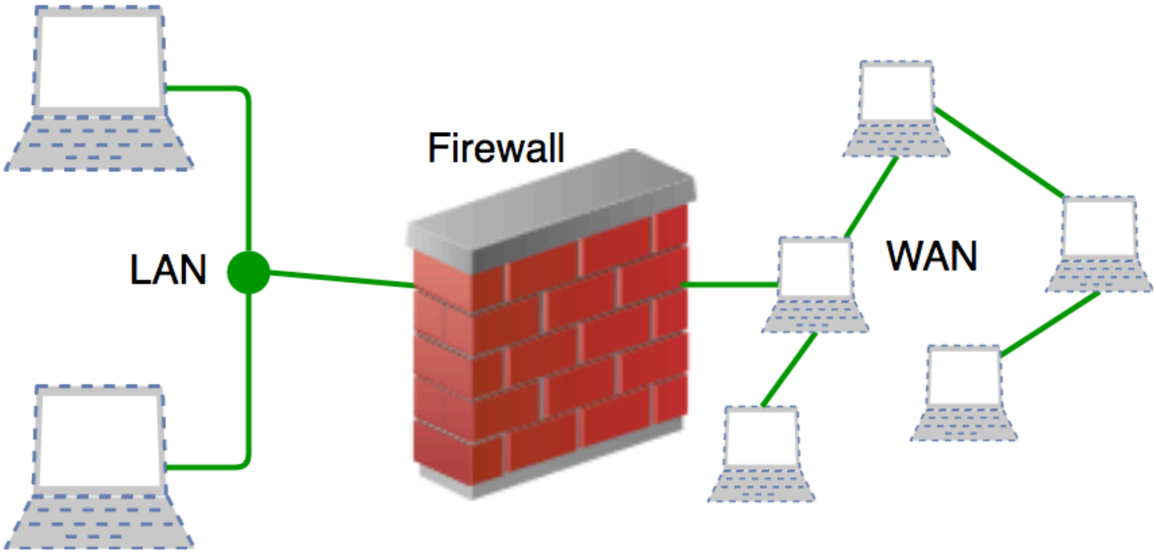
NAT Gateway -

NAT Gateway stands for Network Address Translation. It is a managed AWS service that is scaled based on your usage. NAT Gateway will help you to access the internet which instances are configured in the private subnet but without proper routing, no one can access that instance from outside.

What is Firewall?

A firewall is a network security device, either hardware or software-based, which monitors all incoming and outgoing traffic and based on a defined set of security rules accepts, rejects, or drops that specific traffic.

A firewall is a type of network security device that filters incoming and outgoing network traffic with security policies that have previously been set up inside an organization. A firewall is essentially the wall that separates a private internal network from the open Internet at its very basic level.



Amazon Virtual Private Cloud (VPC) - is a virtual network that allows users to launch AWS resources in a logically isolated section of the AWS cloud. It gives users full control over their virtual networking environment.

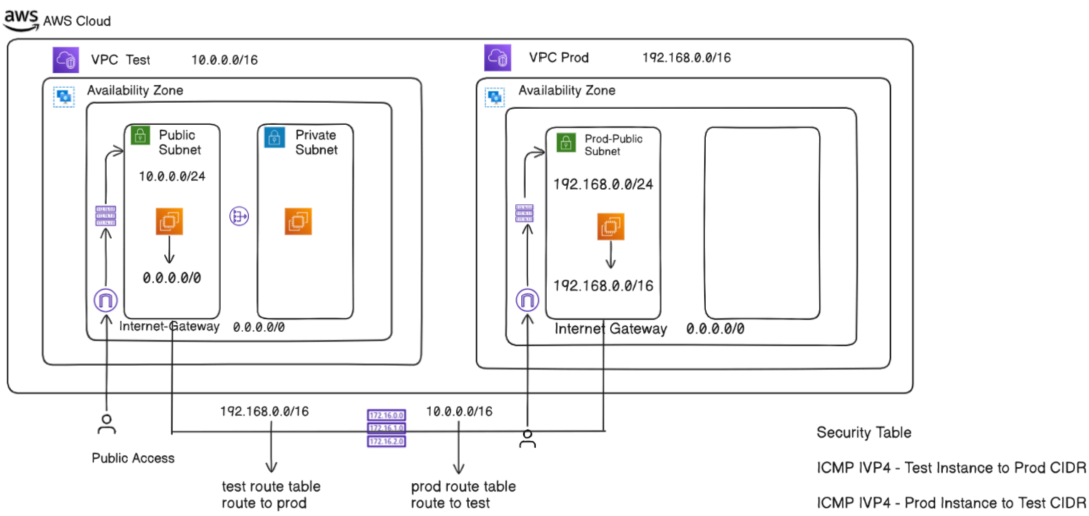
Features

- **IP address ranges**: Users can specify IP address ranges for their VPCs.
- **Subnets**: Users can create subnets, which are ranges of IP addresses within a VPC.
- **Route tables**: Users can configure route tables to control how traffic moves to and from subnets.
- **Network gateways**: Users can configure network gateways to allow communication between VPC resources and the internet.

- **Security groups**: Users can associate security groups with subnets to control access to Amazon EC2 instances.

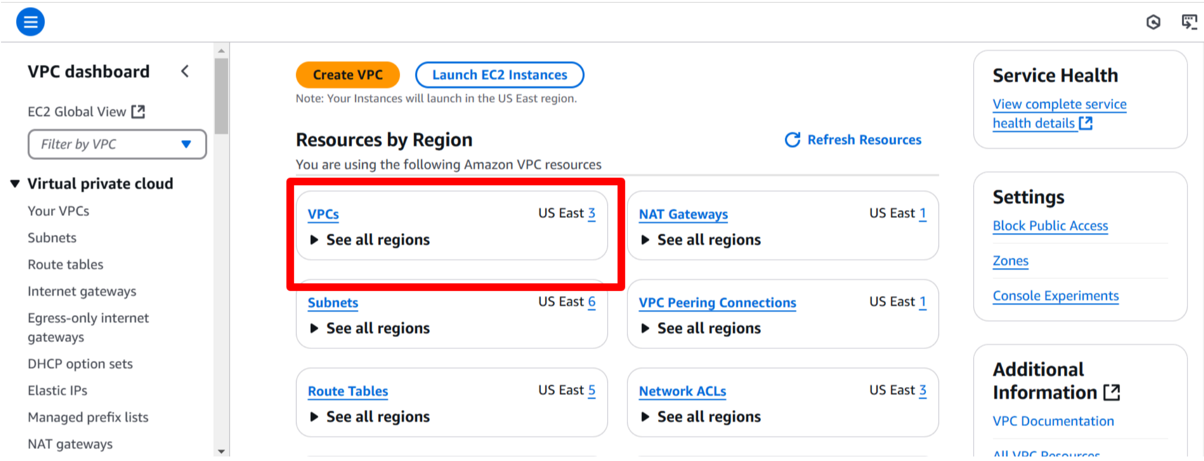
A *virtual private cloud* (VPC) is a virtual network dedicated to your AWS account. It is logically isolated from other virtual networks in the AWS Cloud. You can launch AWS resources, such as Amazon EC2 instances, into your VPC.

A VPC peering connection is a networking connection between two VPCs that enables you to route traffic between them using private IPv4 addresses or IPv6 addresses. Instances in either VPC can communicate with each other as if they are within the same network. You can create a VPC peering connection between your own VPCs, or with a VPC in another AWS account. The VPCs can be in different Regions (also known as an inter-Region VPC peering connection).

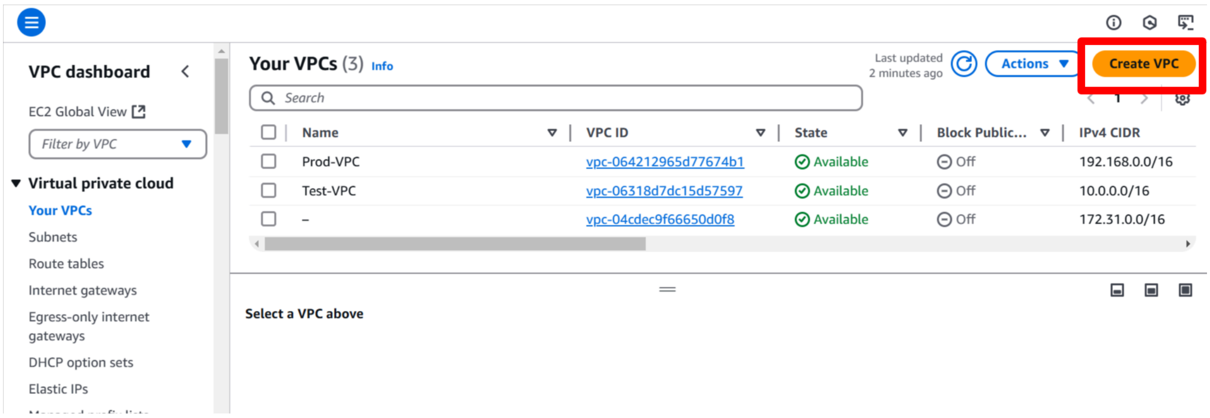


Steps to create VPC in AWS Account -

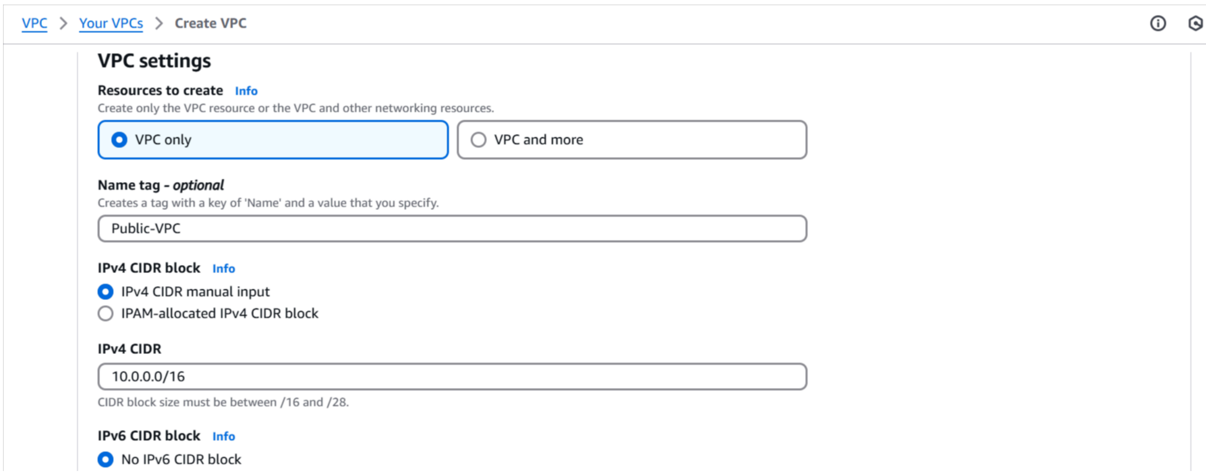
1. Go to the VPC dashboard in your AWS account and select the VPC option.



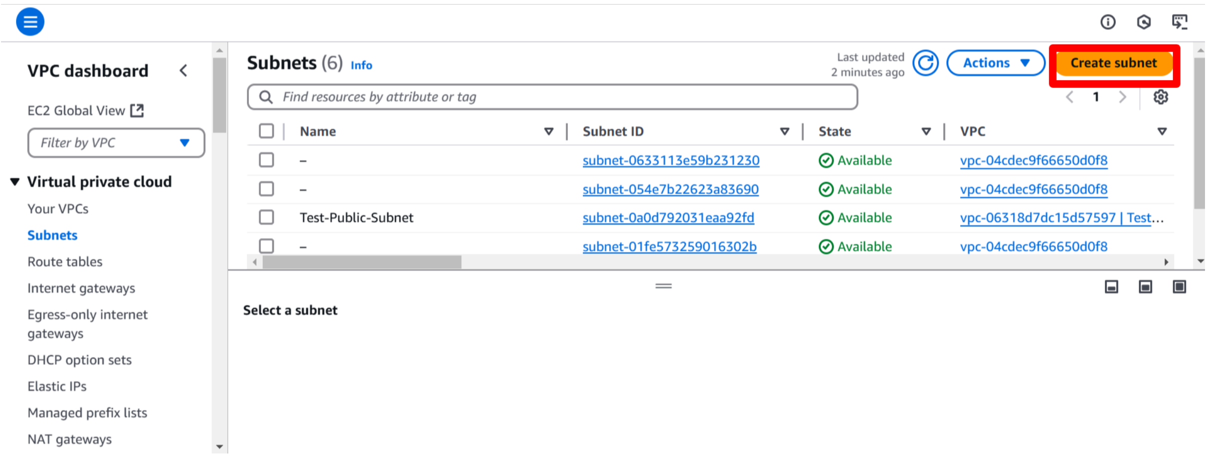
2. Click on the create VPC button to create a new VPC.

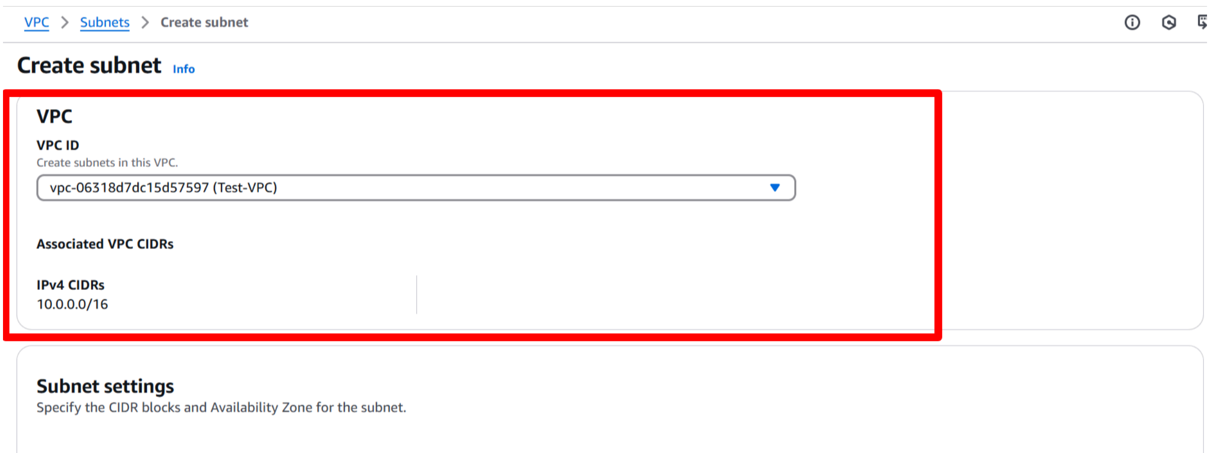


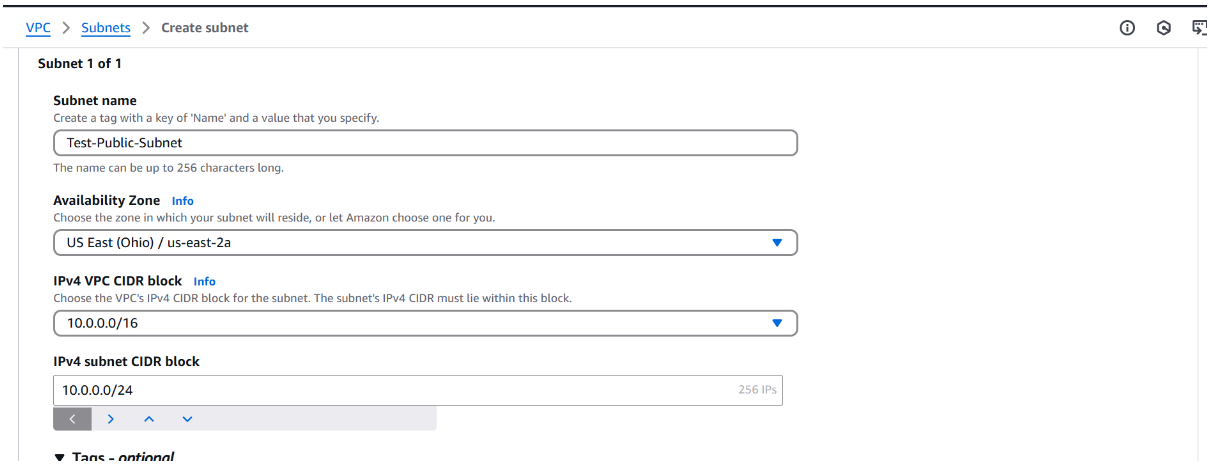
3. Fill out required fields to create the VPC network and click on create VPC.



4. After creating VPC, go to the subnets tab and create subnets and choose the availability zone for subnets and VPC (Test-VPC) and create the subnet.

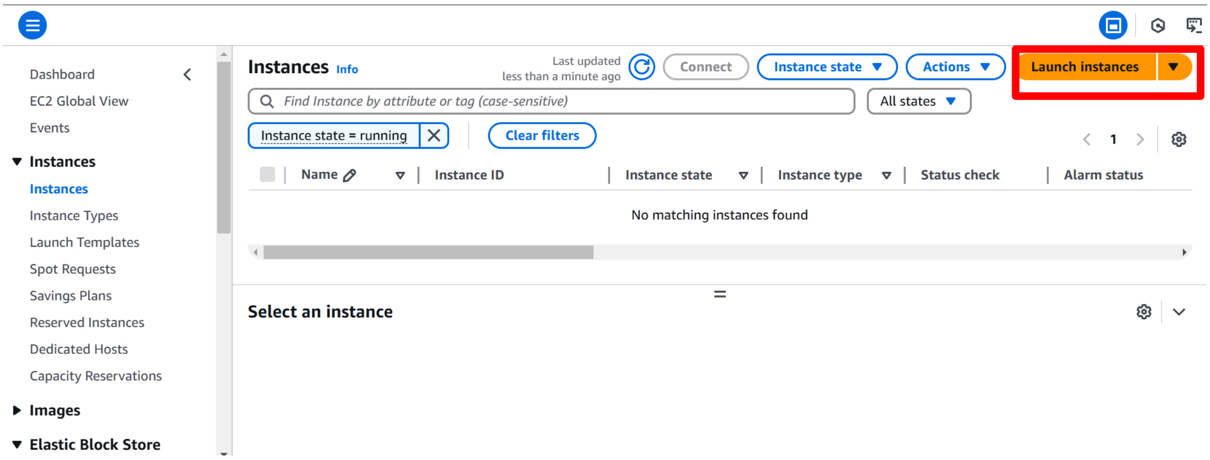


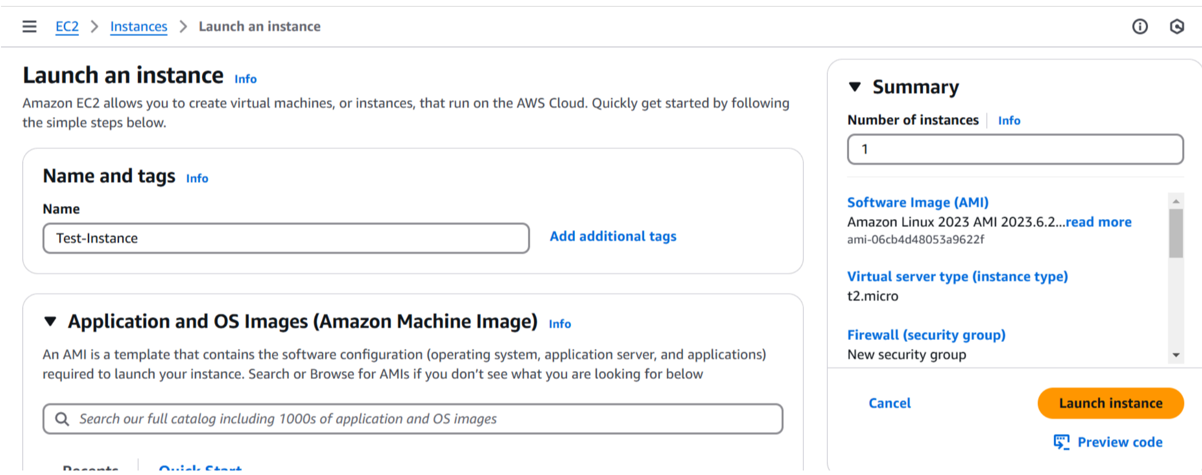




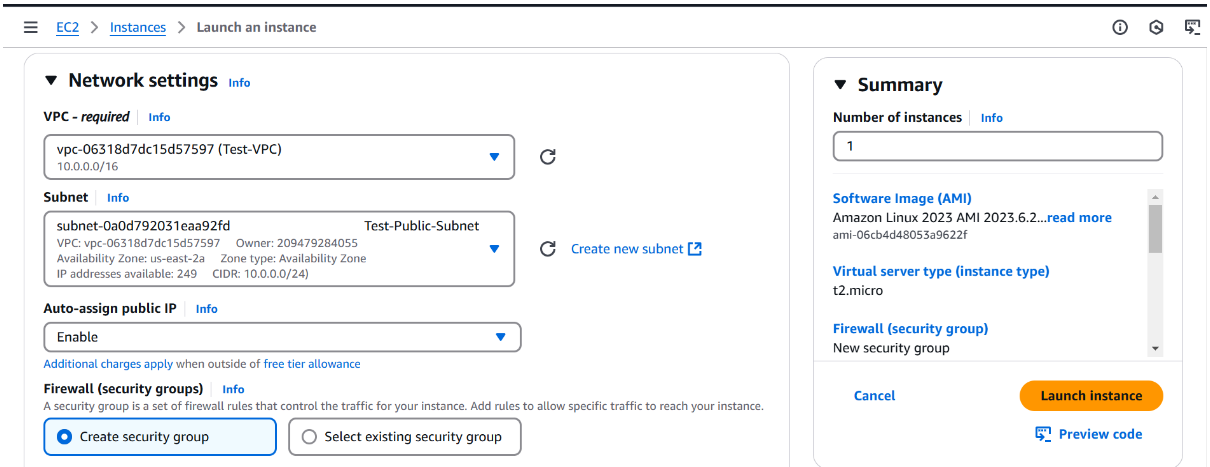
Now, we have created the VPC and subnets for our isolated network.

5. Go to the EC2 tab and create the EC2 instances resources (Test-Instance) inside the subnet and VPC.

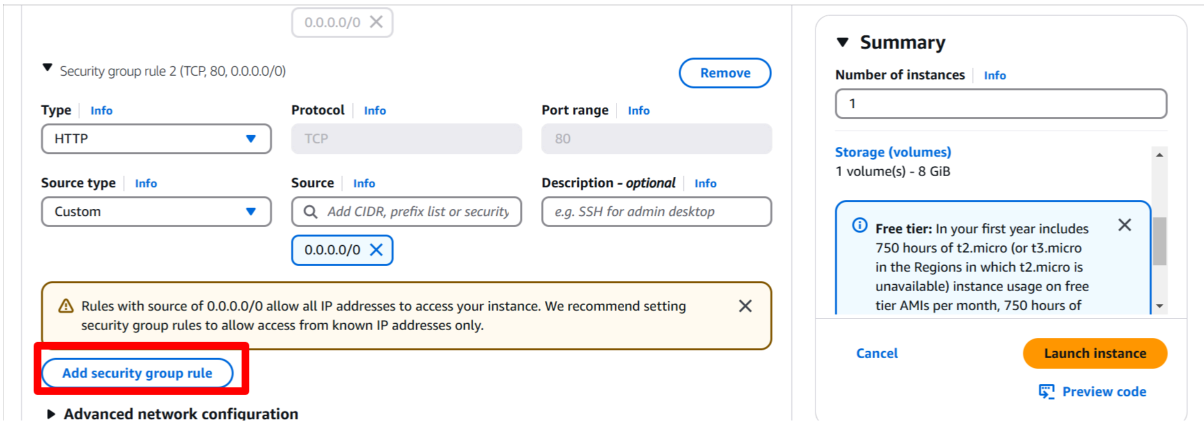




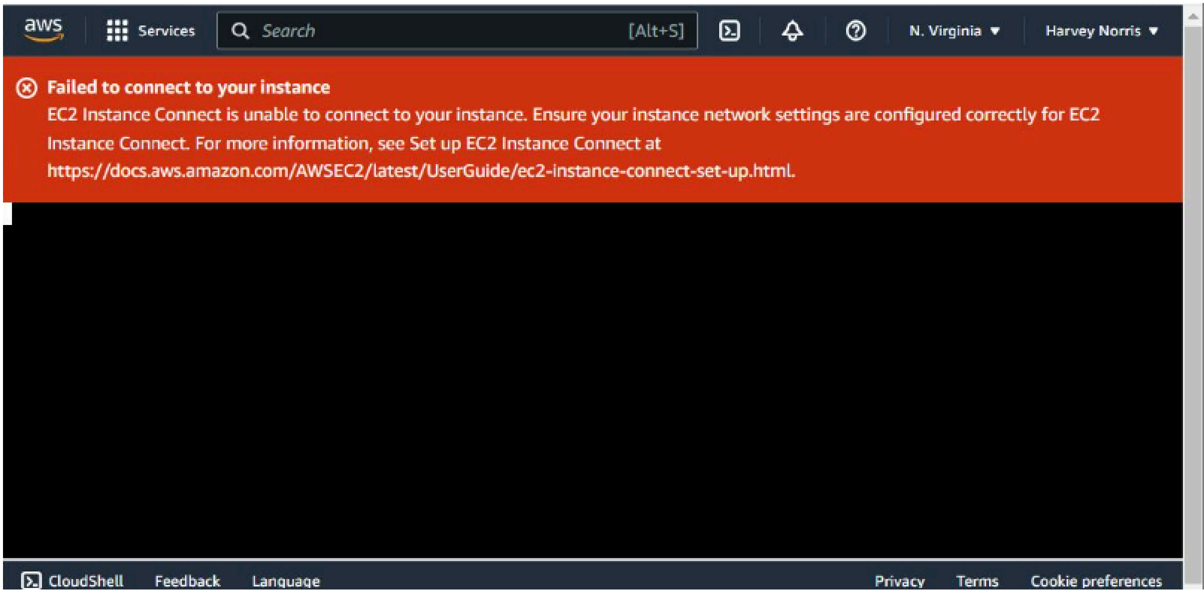
Edit the networking setting of EC2 instances, while creating EC2 and choose our VPC and subnets we have created.



Add Firewalls rule to allow the HTTP protocols so that anyone can access the public resources from anywhere (like the internet) and launch the instance.

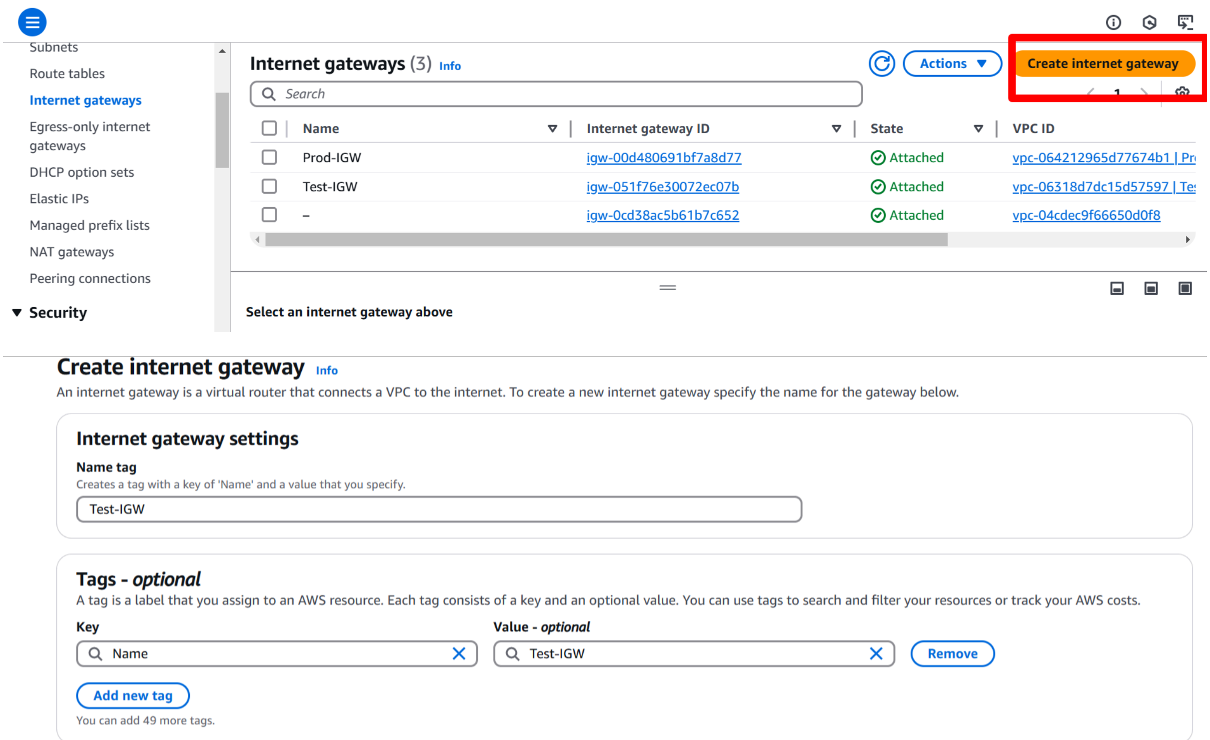


Therefore, we have created the resources inside the VPC, but without an internet gateway no one can access it. If you try to access the instance without attaching the internet gateway instance gives you the error.



Without an internet gateway, you can give access to anyone to access your network and it will become a private network. To allow outside public access attach internet gateway to your public subnet (Test-Public-Subnet)

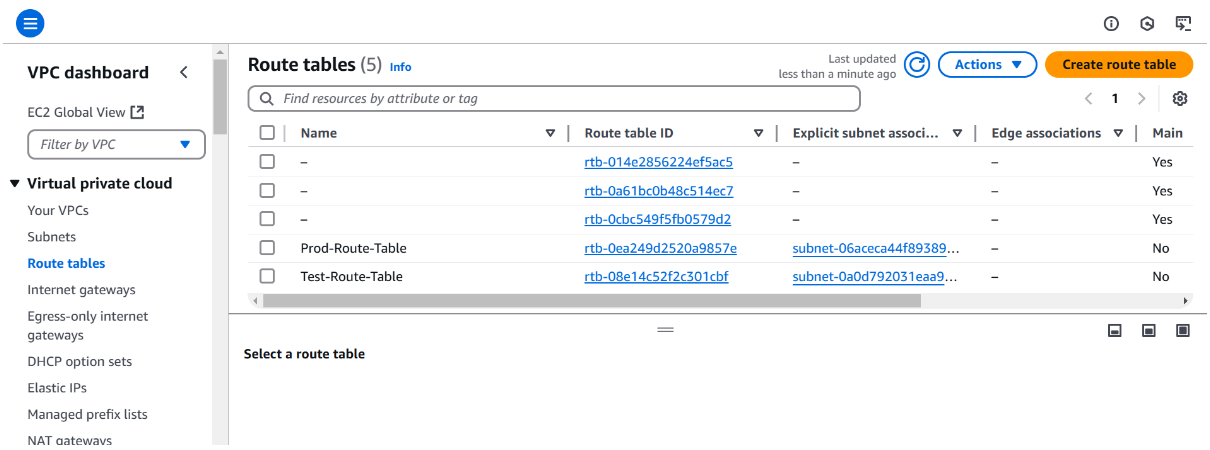
6. Go to the internet gateway tab in VPC service and click on create internet gateway button,



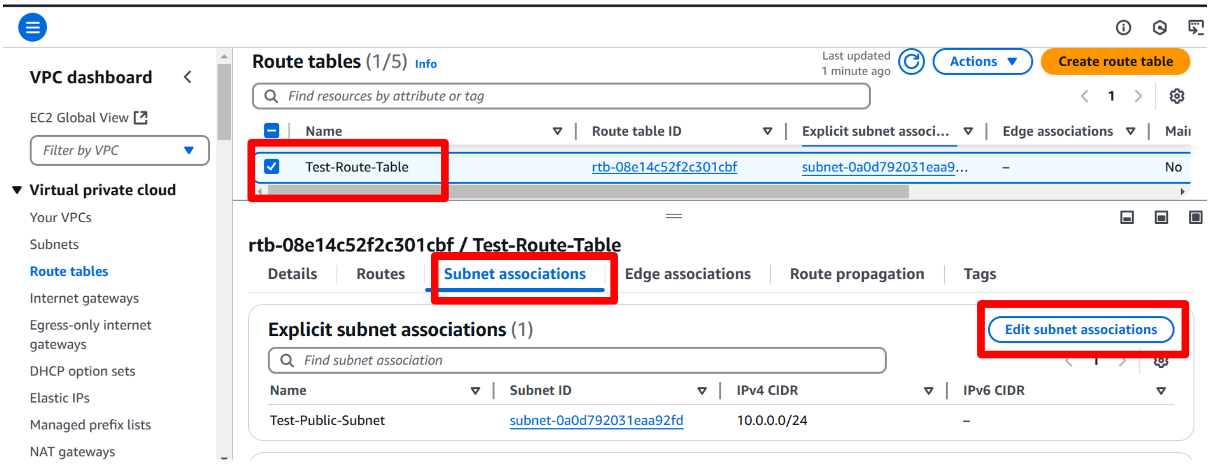
After attaching the internet gateway, attach a route table to your public. This table helps you to route your data to the destination.

Route table is a table where all the IP addresses are stored and this route table attaches to the internet gateway and the subnets directly. That's how we can access the resources inside the public subnets.

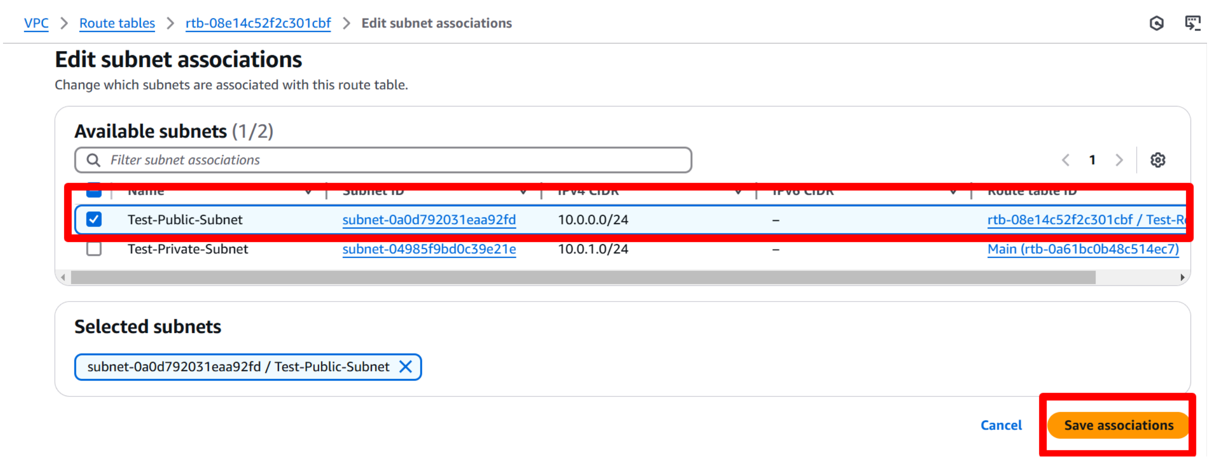
7. Attach the route table to your internet gateway, and provide the rules so that your route table can associate with the public subnets.



Select the Test-Route-Table and go to the subnet association tab edit it to associate your subnet with route table.

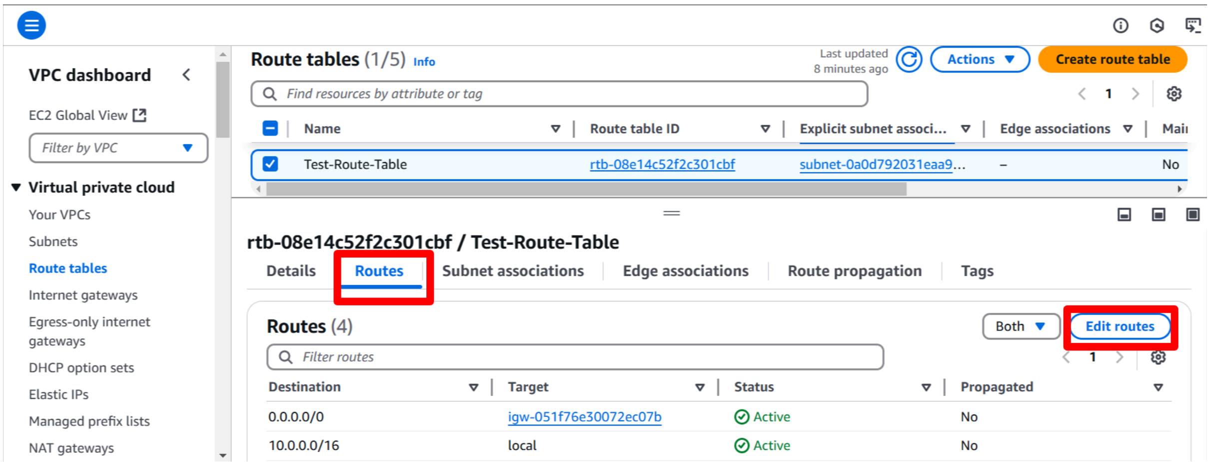


Check the tab of Test-Public-Subnet and associate your subnet.

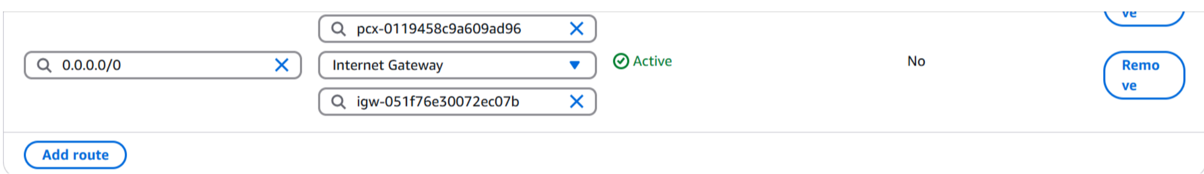


Now the subnet is associated with the route table.

Now we have to give internet access to our instances by adding routes (allow traffic) to the internet gateway.



Now allow public traffic by attaching the internet gateway to routes.



0.0.0.0/0 so that the public can access the resources.

8. Do the same process for creating Prod VPC and add a security group to unique IP addresses so that only few people can access that while creating Prod-Public-Instance.